



Module 23 Day 2

23.3 Student Guide: Cybersecurity Interviewing

Overview

In today's class, we will prepare for the various stages of cybersecurity interviewing. You'll practice common behavioral and technical interview questions and participate in a mock interview group activity.

Class Objectives

By the end of class, you will be able to:

- Prepare for an initial phone interview used to screen candidates for IT and cybersecurity positions.
- Answer technical and behavioral interview questions.
- Conduct, answer, and provide constructive feedback to mock interview questions and answers.

01. Introduction to Cybersecurity Interviewing

Today we will continue our focus on career prep by covering cybersecurity interviews. At this point, you should have a strong resume and a LinkedIn profile that reflects a growing cyber network. Connect with Career Services if you have not yet created or updated your resumes and LinkedIn profiles. You should also have an understanding of how certification can give you an edge in the job market. The next step is to apply for jobs, reach out to key decision makers, and prepare for interviews.

Today, we'll prepare you for interviews by covering the following:

- The purposes of behavioral and technical interviewing and what kind of questions you can expect.
- Mock interviews with behavioral and technical questions.

Previously this week, we learned that:

- Certification is a good way to prove that you have a specific set of skills and knowledge.
 - Certification can provide you with an edge over other job candidates.
 - Cybersecurity professionals take a variety of paths to reach different positions.
 - You can use CyberSeek.com and LinkedIn to help determine paths taken by cyber professionals to reach career goals that are similar to yours.
 - Building a cyber network is one of the best ways to find career opportunities.
 - Non-traditional approaches to job seeking can lead you to less crowded career opportunities.
 - Building a powerful resume and LinkedIn profile can increase your odds of landing an interview and, subsequently, a cyber job.
-

02. Interviewing Tips

As you prepare for a cyber job interview, you should consider the following questions:

- What should I do to prepare for a successful interview?
- What kinds of questions can I expect during the behavioral interview?
- What kinds of questions can I expect during the technical interview?
- What should I avoid during the interview?

First, we'll cover the typical structure of an interview process.

Interview Process

Before receiving a job offer, you can expect to participate in at least three interviews:

1. The initial interview (also known as the phone screener or preliminary interview)
2. The behavioral interview
3. The technical interview

The Initial Interview

Once your resume has been reviewed and you've been selected as a potential candidate, an HR staffer will conduct an initial phone screen interview. This interview aims to:

- Narrow down the list of candidates so that the final decision makers have the best candidates to choose from.
- Provide basic job description, location, and hourly expectations, etc.
- Confirm that the pay range is agreeable for both you and the employer.
- Confirm that this is a position you are interested in.

The staffer may have a few more questions to help them confirm that you are right for the position before they move you to the next interview.

The Behavioral Interview

The behavioral interview is usually next in the interview process.

- A behavioral interview is used by many hiring managers to better understand a candidate's likely future performance by evaluating their communication and soft skills.
- The hiring manager will often use this interview to determine how interested and qualified you are, as well your ability to communicate.
- Behavioral interview questions are often similar across companies.

Below are several common behavioral interview questions and how to answer them.

Question 1: Tell me a bit about yourself.

- This question allows you to:
 - Show your communication skills and accomplishments.
 - Highlight pieces of your personal background, interests, and work experience that show you are right for the job.
 - Briefly explain your work history and how you came to apply to this position.
 - Consider the following tips for answering this question:
 - This question is very common, so be prepared to answer it.
 - Consider using a past, present, future structure in your answer.
 - For example: Your past work history, why you are currently applying to this position, and how you are positioned to be a future contributor in this position.
 - Use this question as an opportunity to express your passion for cybersecurity.
 - Be direct and prepared. Do not ramble. Keep your answer to a few minutes.

Question 2: Why are you interested in this cybersecurity position? Why are you interested in a career in cybersecurity?

- Discuss what led you to this boot camp and what parts were the most interesting or exciting.
- If the question is focused on the exact position, be specific and demonstrate your knowledge of the job description.
 - For example, if the position uses Burp Suite, you might note how you enjoy finding web application vulnerabilities.

Question 3: Why did you leave your last position? Why are you looking for a new position?

- Be careful answering: do not use this question to say negative things about your past employer.
- Instead, stay focused on what you're looking for next, your passion for cybersecurity, and how this position aligns with those things.

Question 4: Why do you want to work for this organization?

- Make sure to prepare this answer, as it demonstrates excitement about working at this company specifically.
- Be specific. For example, point out that you liked the organization's launch of a new malware identification program that was able to identify patterns in ransomware.

Question 5: What questions do you have for us?

- This question will usually come at the end of every interview.
- Not asking any questions can signal to the employer that you lack interest in the job or company.
- Some questions you might ask include:
 - What does a typical day in this position consist of?
 - What is your favorite part of working here?
 - What challenges does your company/department currently face, and how can this position help overcome those challenges?

Summarize this section for students by covering the following tips to prepare for a cybersecurity behavioral interview:

- Be prepared ahead of time to answer the "tell me about yourself" question.
- Research the organization you are interviewing with and be able to note specific facts about the organization.
- Have several questions prepared ahead of time to ask the interviewer.

In the next activity, you will have an opportunity to prepare a response for the "tell me about yourself" question.

03. Activity: Behavioral Interviews

04. Activity Review: Behavioral Interviews

05. Technical Interviews

Once a candidate has successfully moved through a screener and a behavioral interview, they will be invited to a technical interview.

- The technical interview is a common step in the cybersecurity interview process in which candidates demonstrate their technical and problem-solving skills.
- This interview typically contains three types of questions:
 - Basic technical questions
 - Background technical questions
 - Situational technical questions

Basic Technical Questions

Basic technical questions gauge your knowledge of specific concepts and tools. They might include:

- What is the difference between TCP and UDP?
- What is the difference between data in motion and data at rest in terms of security?
- How many keys does asymmetric encryption use?

When answering these questions:

- It is acceptable and recommended to recite back the question before answering.
- Answer the question and provide additional details if you are able.
- If you do not know the answer, be honest and explain that you do not know.
 - In these cases, you can explain how you would research the answer. And as a bonus, you can send the interviewer the answer in a follow-up email.

While it is important to be direct when answering interview questions, you should provide a level of detail that shows comprehensive knowledge.

For example, consider the following basic and detailed answers.

- What is the difference between TCP and UDP?
 - Basic: TCP is connection-oriented and UDP is connection-less.
 - Detailed: TCP is connection-oriented and UDP is connection-less. TCP works well for applications that require high reliability, while UDP works well for applications that prefer higher speed over reliability, such as video games or video streaming.
- What is the difference between data in motion and data at rest in terms of security?
 - Basic: Data in motion is data that is transferred on the network between machines, or "live on the wire." Data at rest is data that is being stored on a static file system, such as a hard drive or database.
 - Detailed: Data in motion is data that is transferred on the network between machines, or "live on the wire." Data at rest is data that is being stored on a static file system, such as a hard drive or database. Encrypting data in motion typically requires faster algorithms than encrypting data at rest. When data is transferred between machines, slow algorithms manifest as high latency (i.e., long transfer times). Reducing latency requires using

faster algorithms. For this reason, protocols like SSL/TLS use symmetric encryption for the main data transfer and avoid slower asymmetric methods.

- How many keys does asymmetric encryption use?
 - Basic: Two.
 - Detailed: Two. Asymmetric encryption uses a private key and a public key. Each individual is required to have this two-key pair.

Background Technical Questions

Background technical questions are used to gauge a candidate's exposure to and experience with real-world scenarios.

Consider the following example:

- Describe a breach or security vulnerability that you contained.
 - Remember, it is acceptable and recommended to recite the question back before answering.
 - If you have experience containing a breach of a security vulnerability, explain your specific process step-by-step.
 - If you do not have professional experience, pick a breach or vulnerability from the news and discuss how you would contain it.
 - For example, "XYZ organization didn't patch their Apache servers, exposing an Apache Struts vulnerability. I would ensure that our organization had a detailed process to guarantee all systems were patched appropriately and in a timely manner."

Situational Technical Questions

Situational technical questions gauge your decision-making ability when presented with a problem.

Consider the following question:

- Suppose you've implemented a firewall policy on an HTTP server. How would you test it?
 - First, restate the question:
 - "So, in this situation, I have a host running an HTTP server and I've set some firewall rules to allow access only to specific ports and block attackers from all others."
 - Next, provide elaboration and a conceptual solution:
 - "Since this is an HTTP server, I guess you only want to allow access to/from port 80 and 443. If the firewall is working properly, I should be able to send an HTTP or HTTPS request to the server from a foreign host and get a response back. I should not be able to get a response from any other port."

- Finally, explain the technical steps and tools needed to implement your solution:
 - "To test that I'm getting HTTP and HTTPS requests/responses, I'd use curl to hit the firewalled server from a foreign host. I'd expect to get a response. To test that all the other ports are closed, I'd use Nmap to run a port scan."
-

06. Activity: Technical Interviews

07. Activity Review: Technical Interviews

Technical interviews are a vital opportunity in the hiring process to prove your knowledge of key skills and tools.

Even if you are unable to speak first-hand about professional experiences, you can demonstrate a proactive and passionate approach to problems you haven't experienced yet.

08. Break

09. Introduction to Mock Interviews

The best way to prepare for job interviews is to practice. The rest of class will be spent conducting mock interviews.

- You will work in groups of four or five to conduct, answer, and observe mock interviews.
- To simulate a real job interview, you will be asked behavioral and technical questions. You will not know which will be asked.
- There will likely be questions you do not know the answer to. How you respond to those questions is very important.
- Everyone in the group will have an opportunity to be interviewed and to conduct the interview.

Everyone should have a copy of their resume in front of them as well. In interviews, you will likely be asked questions about your resume.

10. Activity: Mock Interview

11. Activity Review: Mock Interview

The best way to prepare for these interviews is to practice them.

- You can use note cards to practice every type of interview question and answer.
- Candidates are usually given a few days to a week to prepare for an interview. You should take advantage of that time to prepare. Cramming the night before is not the best approach.

Review the following resources for examples of cyber interview questions:

- [Daniel Miessler: 60 Cybersecurity Interview Questions](#)
- [Guru99: Top 100+ Cyber Security Interview Questions and Answers](#)
- [Edureka: Top 50 Cyber Security Interview Questions and Answers 2024](#)
- [Indeed: 111 Popular Cyber Security Interview Questions and Answers](#)

[<- Back to Module Contents](#)
